Why Bitter Coffee Starts to Taste Better

From: ScienceDaily Strange & Offbeat Mined: 2026-11-10 3 min read

Your first sip of black coffee can taste awful. Our brains are wired to avoid bitterness, yet millions of people grow to love that taste. Scientists at UC San Francisco may have found part of the answer.

The team studied **dopamine**, a chemical messenger for motivation, reward and learning. This study, published November 4 in Nature, looks at a slow **dopamine** system that responds to news from inside the body.

When food reaches the gut, sensors spot **nutrients** such as sugar and fat, and the brain can answer with a **dopamine** response that builds slowly and lasts several minutes.

Knight and Grove gave mice sugar water or a milkshake-like liquid straight into their stomachs. The strongest response showed up in the **anterior basolateral amygdala**, or BLA, a brain area for emotion and learning.

Blocking the pathway left mice eating normally, but they could no longer learn that a new flavor meant **calories** were coming. When the team switched the fibers on, mice learned to like a flavor with no **calories**. Once the brain links coffee's taste with **caffeine**'s lift, Knight said, coffee tastes a lot better.

Word Treasure

dopamine -- Dopamine is a chemical messenger in the brain that helps with motivation, reward and learning.

nutrients -- Nutrients are the useful parts of food, like sugar and fat.

anterior basolateral amygdala -- The anterior basolateral amygdala is a brain area for emotion and learning.

calories -- Calories show how much energy food gives the body.

caffeine -- Caffeine is the part of coffee that gives people a lift.

Background you need

Dopamine is a chemical messenger in the brain that helps with motivation, reward and learning.

When food reaches the gut, sensors there can spot nutrients, and the brain can slowly respond with dopamine.

Over time the brain can learn to link a flavor with the good feeling that follows, so a taste people once disliked can start to seem pleasant.

Break it down

WHO: Scientists at UC San Francisco, including Knight and Grove, and the mice they studied

WHAT: A slow dopamine system that helps the brain learn to like flavors linked to calories or caffeine's lift

WHERE: Inside the body, in the gut and in a brain area called the anterior basolateral amygdala

WHY: To explain why people grow to love bitter tastes like black coffee even though brains are wired to avoid bitterness

Why it matters

It helps explain why foods you once disliked can start to taste good after you try them a few times, just like a new food at school lunch.

Test what you remember

1. What is dopamine, as described in the story?

(A) A kind of sugar found in coffee
(B) A bitter flavor in food
(C) A chemical messenger for motivation, reward and learning
(D) A part of the stomach

2. Where did the strongest dopamine response show up in the mice?

(A) In the anterior basolateral amygdala, or BLA
(B) In the tongue
(C) In the stomach
(D) In the eyes

3. What happened when the pathway was blocked?

(A) Mice could no longer taste anything
(B) Mice could no longer learn that a new flavor meant calories were coming
(C) Mice liked every flavor at once
(D) Mice stopped eating

4. What happened when the team switched the fibers on?

(A) The mice refused all food
(B) The mice forgot every flavor
(C) The mice fell asleep
(D) Mice learned to like a flavor with no calories

5. Why does coffee start to taste better, according to Knight?

(A) Because coffee beans change color
(B) Once the brain links coffee's taste with caffeine's lift
(C) Because the brain gets used to the heat
(D) Because coffee loses its caffeine

6. What do sensors in the gut spot?

(A) The temperature of the room
(B) Colors and shapes
(C) Nutrients such as sugar and fat
(D) Sounds and music

Answer key (try the quiz first!): 1: C 2: A 3: B 4: D 5: B 6: C

Questions to think about

MY THOUGHTS (at least 20 words):